

CCNA 200-301 v1.1 Master Cheat Sheet

Based on the uploaded Cisco CCNA Exam v1.1 (200-301) blueprint. Organized in the same order as the official objectives. This is a concise study guide, not an exhaustive textbook.

1. Network Fundamentals

- Know: IPv4/IPv6 addressing, subnetting, TCP vs UDP, MAC learning, ARP, wireless basics, virtualization, VRFs.
- Commands: show ip interface brief; show ipv6 interface brief; show arp; show mac address-table; ping; traceroute.
- Must know: private IPv4 ranges, IPv6 types, default gateway, longest prefix match.

2. Network Access

- VLAN: vlan ; name NAME; interface; switchport mode access; switchport access vlan X; show vlan brief.
- Trunks: switchport mode trunk; switchport trunk native vlan X; show interfaces trunk.
- EtherChannel (LACP): channel-group X mode active/passive; show etherchannel summary.
- STP: spanning-tree vlan X root primary; spanning-tree portfast; spanning-tree bpduguard enable; root guard; loop guard; bpdufilter.
- Verify: show spanning-tree.

3. IP Connectivity

- Routes: ip route NETWORK MASK NEXT-HOP; ipv6 route PREFIX NEXT-HOP.
- OSPF: router ospf 1; network A.B.C.D WILDCARD area 0; passive-interface; router-id.
- Verify: show ip route; show ip ospf neighbor; show ip ospf interface brief.

4. IP Services

- NAT, DHCP, DNS, NTP, SNMP, Syslog, SSH, QoS, FTP/TFTP.
- SSH: hostname; ip domain-name; crypto key generate rsa; username; line vty; transport input ssh.
- Verify: show ip nat translations; show ntp associations; show ip ssh.

5. Security Fundamentals

- ACLs: access-list / ip access-list; permit/deny; apply with ip access-group.
- Port Security: switchport port-security; maximum; violation; mac-address sticky.
- DHCP Snooping, DAI, AAA, VPN concepts, WPA2/WPA3.
- Verify: show access-lists; show port-security interface.

6. Automation & Programmability

- REST APIs, JSON, SDN, controllers, northbound/southbound APIs, Ansible, Terraform, AI/ML concepts.
- Know CRUD and HTTP verbs: GET, POST, PUT, PATCH, DELETE.

High-yield show commands

show running-config
show startup-config
show interfaces
show ip interface brief
show vlan brief
show interfaces trunk
show spanning-tree
show etherchannel summary
show ip route
show access-lists